CSC 4500 Database Design and Implementation
Homework #3 Due Thursday, November 12, 11:59 pm

For this assignment, you will work in a team (Team assignments in GitHub) to complete this assignment. You and your team will work together to construct a mini database project (SCHEMAS to choose from in GitHub) that includes a video presentation. Failure to attend class and actively engage with your group will negatively impact your grade.

1. Get together with your group to select a schema you will be working on. Email me once you have selected one. It is on first come first serve basis.

2. Create the database, table and insert values (from a .csv file) to build your database in MySQL. Your tables must use appropriate attribute names, datatypes, foreign key constraints and index structures. Ensure you have enough rows for each of the entities.

3. Write a Java Swing/AWT and JDBC programs using a menu driven interface where the action of the code will consist of either insert or update statements to be used directly as inserts to help populate your tables.

4. Write a Java Swing/AWT and JDBC program that will run your queries (see query requirements below) using a menu driven interface for the user to choose which query to execute. The output of each query should be displayed in a user-friendly tabular format. Pay attention to the user input requirement stated below. Your code must be well structured, efficient, modularized, and correct.

5. Create a presentation where each group member presents their work for a technical and non-technical audience. This video must flow well between group members.

Queries:
Groups of 3: Must include 6 different queries.
Groups of 4: Must include 8 different queries.

Each group member is responsible for

- Contribute to the ER model design, create table and insert statements from the .CSV files.
- At least one GUI input and insert statement from a JDBC program to insert into a table.
- At least one query where it must include user input when executed in the JDBC program
- At least one query where it must include data from at least two tables.
- Your queries must include an example of a division style query, subquery or corelated subquery, a group by clause, and a case statement.
- 2-3 minutes of video content where the group member presents their work – remember, for a non-technical audience

What to submit?

Each group member uploads

- A .docx and .sql file of their queries written to run in MySQL Workbench with screenshots demonstrating they work. Before each screenshot, you must include a single sentence that explains what your query does in non-technical terms. In addition, you must explain your query in technical terms. In other words, include two complete sentences describing each query: one sentence written for a technical audience and one written for a non-technical audience.
- A **.java** file that contains the Java Swing/AWT and JDBC code.

One group member uploads

- A .sql file that contains the create table and create/insert statements
- A .docx file that contains the technical and non-technical description of the insertions and queries
- A .java file that contains the Java Swing/AWT and JDBC code
- The group video for a non-technical audience.